Power Savings

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Power, cooling, and space requirements are some of the key challenges faced by IT managers today. The Environmental Protection Agency has predicted that datacenter power usage will double over the next five years. IT analysts believe that by the end of the decade the world's datacenters will have run out of power. Environmental protection issues are gaining much more visibility in the enterprise today with an increasing number of datacenter managers losing sleep over the matter. This has brought the focus on green computing in the datacenters.

While datacenters remain an essential part of most businesses, there will come a time very soon when growth may need to be limited because of power costs and the growing awareness at governmental, corporate, and society on the environmental impact of excess energy consumption. The growing restrictions on power, cooling, and use of datacenter footprint can be seen as coming into conflict with another major IT trend namely a steady increase in the percentage of disk in the data protection mix. Using disk as part of a backup strategy is growing rapidly as IT departments take advantage of the faster backup and restore performance that disk delivers. In turn, disk provides higher service levels to their customers and reduces the use of removable media. But disk storage is fast becoming one of the primary power and cooling culprits in the datacenter, and when disk is used for backup, the requirements can increase dramatically.

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The solution is not to stop using disks for data protection, but to take advantage of various storage technologies by combining different kinds of media for an optimal balance of performance and cost and energy consumption. A combination of data-duplication and a multi-tiered storage architecture can be a key enabler of a future to green computing.

Data De-duplication Makes Sense

Data de-duplication reduces the amount of disk required to protect a given amount of primary data by 90 per cent or more by detecting and eliminating redundant blocks within files and, in some products, between different files and file types. It allows users to exploit the performance characteristics of disk for backup without incurring the costs of conventional disk systems – costs which include both capital expenses and the operating expenses that include power and cooling.

Disaster-recovery protection also benefits from data de-duplication. Using the same technology that identifies duplicate segments inside datasets, data de-duplication systems also reduce the bandwidth needed to transmit backup sets over a network. With data de-duplication, once systems are synchronised, whole backup sets can be replicated while only changed blocks are actually moved. If a new backup is only 5 per cent different from a previous one at a block level, then we can reduce the bandwidth needed for transmission by up to 95 per cent. Replication allows multiple copies of backup data to be maintained on disk in different sites using WAN links. Its use can simplify backup and reduce costs by decreasing an IT department's need to buy, manage, and move tapes.

A Green Tomorrow

The optimal approach to data protection and retention is to match the various business requirements with the right storage technology. IT managers should consider the business requirements for data access and retention, and balance those against the requirements for power, space, and cooling. A multi-tier storage architecture that includes a mix of disk, data de-duplication, and tape technologies could be the ideal solution to drive cost optimisation and enhance energy efficiency.

Datasets with high access requirements may need to use conventional disk systems without data de-duplication. However, they are only practical for a subset of data where the recovery time objective overrides other considerations such as long-term retention and total cost. Datasets with normal access requirements can use disk with data de-duplication technology both as first site for backup data and for medium term retention. Data de-duplication technology reduces space, power, and cooling requirements enough to make it practical to use disk as a retention medium for weeks or months without breaking the operational budget. However, as data needs to be retained for multiple quarters and years and the demand for fast recovery diminishes, the most effective retention medium for most users is tape. Tape cartridges in a tape library consume power at lower rates than any disk system, and tape cartridges stored in a vault consume the least of all, as well as providing the lowest aggregated storage cost per GB.

A multi-tiered storage architecture with the appropriate mix of disk, tape, and enabling technologies like data de-duplication could be the appropriate solution to address key concerns around datacenter power, cooling, and space requirements. A multi-tiered storage architecture would help attain operational efficiencies while reducing power consumption and enable green computing. ■

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